LIVING THINGS

# Humans and the Environment

How people help and harm ecosystems, protecting animals and plants, and caring for the environment



recycling



reusable bag



deciduous tree



solar energy

**In this pack**

- **Learn it:** People and our environment ★☆☆
- **Fact check:** Helping or harming? ★☆☆
- **Think it through:** Problems and solutions ★★☆
- **Investigate:** Our school litter survey ★★☆
- **Real-world case study:** The golden fibre: can jute beat plastic? ★★★
- **Review and challenge:** Show what you know ★★★

Also covers Cambridge Primary Science Stage 4: Unit 2 Living things and environments (2.6 How we affect the environment, 2.8 Recycling).

**This pack belongs to:** _____ **Class:** _____

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# People and our environment



Name: Class: Date:

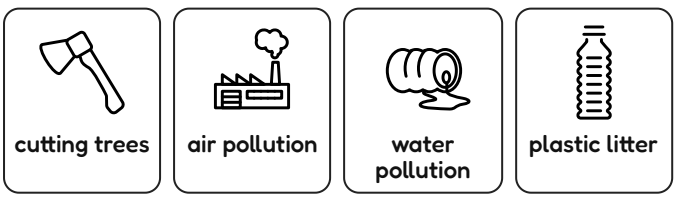
**Learning goals**

- Describe good and bad effects people have on the environment.
- Explain how human activities affect animals and plants in Bangladesh, and suggest solutions.
- List ways to care for the environment: reduce, reuse, recycle and more.

People are part of many ecosystems. We take water from rivers, grow food in fields, cut wood from forests and catch fish from the sea. Long ago, there were far fewer people and they used simple tools. Today there are more than 170 million people in Bangladesh, and our actions can help or harm the living things around us.

## How we can harm the environment

- **Cutting down forests** (deforestation) destroys the homes of animals such as elephants and gibbons.
- **Pollution**: rubbish, plastic, smoke and dirty water from factories harm living things. Rivers near cities, such as the Buriganga, have become very polluted.
- **Hunting and overfishing** can make animals rare. Catching too many young fish leaves none to grow up and breed.
- **Filling in wetlands** to build houses and roads destroys the homes of fish, frogs and birds.




When a living thing loses its habitat it may die out from that place. If it dies out everywhere, it is **extinct** and can never come back. Many kinds of animals and plants are now **endangered** (in danger of becoming extinct).

## How we can help the environment

- **Protecting habitats**: national parks and wildlife sanctuaries, such as the Sundarbans and Lawachara.
- **Planting trees**: in Bangladesh, people plant millions of trees every year during the monsoon.
- **Laws** that stop hunting, protect fish while they breed, and ban harmful things such as thin plastic bags.
- **Cleaning up** rivers, beaches and parks.

## Caring for the environment every day

| Action | What it means | Example |
|---|---|---|
| **Reduce** | use less | take a cloth bag to the market |
| **Reuse** | use things again | use a jar to store rice; give old clothes to others |
| **Recycle** | make used materials into new things | collect paper, glass, metal and plastic for recycling |
| **Save energy** | use less electricity and fuel | switch off lights and fans when you leave a room |
| **Don't litter** | put rubbish in a bin | never throw plastic into drains or rivers |


reduce


reuse


recycle


switch off

**Did you know?**

A plastic bottle can take **hundreds of years** to break down. Plastic in rivers flows to the sea, where turtles and dolphins can mistake it for food.

## Encouraging others

One person can make a difference, but many people together can make a big difference. You can talk to your family, make posters, start an eco-club at school or join a clean-up day.

**Bangladesh connection**

In **2002**, Bangladesh became one of the first countries in the world to **ban thin plastic shopping bags**, after they blocked drains and made floods worse in Dhaka.

**Try this at home**

Do a "rubbish diary" at home for one day. Write down everything your family throws away. Which things could be reduced, reused or recycled?

**Key words**

| | |
|---|---|
| **environment** | everything around us: air, water, land and living things |
| **pollution** | harmful things put into the environment |
| **deforestation** | cutting down forests |
| **extinct** | no longer living anywhere on Earth |
| **endangered** | in danger of becoming extinct |
| **reduce** | use less |
| **reuse** | use again |
| **recycle** | make used materials into new things |

FACT CHECK · A6 · HUMANS AND ENVIRONMENT

# Helping or harming?

★☆☆ Grade 4

Name: Class: Date:

**Learning goals**

- Sort human actions.
- Use reduce, reuse and recycle.

## 1 Sort the actions. [6]

SORT THESE

**using a cloth bag** **planting trees** **throwing plastic in a river** **leaving lights on** **cleaning a beach** **hunting wild birds**

| Helps the environment | Harms the environment |
|---|---|
| | |
| | |
| | |

## 2 Write reduce, reuse or recycle for each action. [4]

| Action | Reduce, reuse or recycle? |
|---|---|
| taking a water bottle from home instead of buying one | |
| using an old tin as a pencil pot | |
| taking newspapers to the paper collector (*bhangari*) | |
| printing on both sides of the paper | |

## 3 Fill in the gaps. [4]

WORD BANK

**deforestation** **extinct** **habitat** **pollution** **sanctuary** **recycling**

Cutting down forests is called _____. An animal that has died out everywhere is _____. Harmful things in the air or water are called _____. A protected forest for animals is a wildlife _____.

# Problems and solutions

★★☆
Grade 4

Name: Class: Date:

**Learning goals**

- Explain human impacts on living things.
- Suggest solutions.

**1** **For each problem, explain how it harms animals or plants and suggest a solution.** [6]

| Problem | How it harms | Solution |
|---|---|---|
| forests cut down for farms | | |
| factories pour dirty water into rivers | | |
| too many young fish are caught | | |

**2** **Long ago, people in Bangladesh carried shopping in jute bags and baskets. Today many use plastic. How has this change affected the environment?** [2]

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.....

.....

**3** **Why is it important to tell others about caring for the environment?** [2]

.....

.....

INVESTIGATE · A6 · HUMANS AND ENVIRONMENT

# Our school litter survey

★★☆
Grade 4

Name: Class: Date:

**Learning goals**

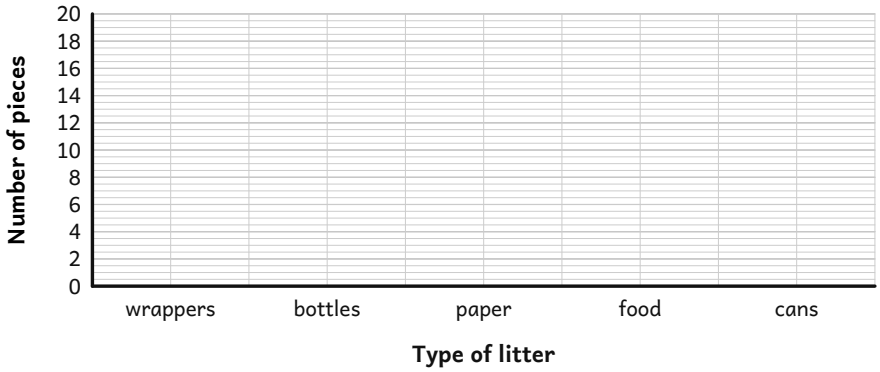
- Collect and tally data.
- Draw a bar chart.
- Plan action from evidence.

Class 4A wore gloves and collected litter from the school playground at the end of one day. They sorted it and counted each type.

**1** **Complete the tally chart.** [2]

| Type of litter | Tally | Total |
|---|---|---|
| sweet wrappers | IIII IIII IIII III | |
| plastic bottles | IIII II | |
| paper | IIII IIII | |
| food waste | IIII | |
| cans | II | |

**2** **Draw a bar chart.** [3]




**3** **Which type of litter was most common? Suggest why.** [2]

.....

.....

**4** **Use the evidence to plan two actions for the school.** [2]

**5** **Why did they wear gloves?** [1]

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# The golden fibre: can jute beat plastic?

★★★ Grade 4

**Name:** **Class:** **Date:**

**Learning goals**

- Compare materials for the environment.
- Use numbers to compare.
- Weigh up different views.

Science · History · Economics · Design · Environment

**Jute** is a plant that grows well in the warm, wet fields of Bangladesh. Its long stems give a strong fibre that is made into bags, sacks, rope and mats. It was once so valuable that it was called the **golden fibre**, and Bangladesh is still one of the world's biggest growers.

Plastic bags are cheap and light, but they can last for hundreds of years and block drains and rivers. Jute bags rot away naturally in a few months, and jute plants take in a lot of carbon dioxide as they grow. A Bangladeshi scientist, Dr Mubarak Ahmad Khan, has developed a bag made from jute that looks like plastic, called the **Sonali bag**.

Based on information about jute in Bangladesh and the Sonali bag. Numbers below are rounded examples.

**1** **Compare a plastic bag and a jute bag.** [4]

| | Plastic bag | Jute bag |
|---|---|---|
| made from | | |
| time to break down | | |
| can it be reused many times? | | |
| cost | | |

**2** **Maths. A family uses 5 plastic bags a week. How many is that in a year (52 weeks)? How many could one jute bag replace?** [2]

.....

.....

**3** **What might each person think about using jute bags?** [3]

| Person | View |
|---|---|
| a jute farmer | |
| a shopkeeper | |
| a fish in the river | |

**4** **Design a campaign to get your school to use jute or cloth bags. Write a slogan and two ideas.** [2]

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# Show what you know

★★★ Grade 4

Name: Class: Date:

**Learning goals**

- Use key vocabulary.
- Answer test-style questions.

## 1 Find the key words.

| | | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|---|
| Y | U | R | E | C | Y | C | L | E | Y | K |
| R | E | D | U | C | E | B | F | F | V | J |
| I | Y | B | F | G | J | W | K | H | O | E |
| R | E | U | S | E | A | Y | L | M | C | R |
| T | V | P | L | A | S | T | I | C | R | E |
| R | O | D | Y | G | Y | K | T | Y | K | W |
| J | E | X | T | I | N | C | T | J | C | T |
| H | A | B | I | T | A | T | E | U | H | D |
| L | D | F | D | N | R | H | R | T | H | C |
| I | K | F | O | R | E | S | T | E | D | V |
| P | G | Y | J | U | M | I | H | S | S | E |

**extinct forest habitat jute litter plastic recycle reduce reuse**

## 2 Which action reduces waste? [1]

- ◯ throwing old toys away
- ◯ buying drinks in small plastic bottles
- ◯ taking a cloth bag to the shop
- ◯ burning rubbish

## 3 An animal that is in danger of dying out is... [1]

- ◯ extinct
- ◯ endangered
- ◯ migrating
- ◯ a producer

## 4 Draw a poster to stop people throwing plastic into rivers. [3]

My poster

**5** **Challenge. Give one way people have helped an ecosystem in Bangladesh and one way people have harmed one. Explain both.** [3]

.....

.....

.....

.....

**How am I doing? Colour or tick a circle.**

| I can... | Not yet | Nearly | Got it! |
|---|---|---|---|
| give examples of people harming and helping the environment | ○ | ○ | ○ |
| explain what extinct and endangered mean | ○ | ○ | ○ |
| use reduce, reuse and recycle | ○ | ○ | ○ |
| carry out a litter survey | ○ | ○ | ○ |
| suggest solutions to environmental problems | ○ | ○ | ○ |
| compare jute and plastic bags | ○ | ○ | ○ |